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Inventor(s)	Ryon; Jason A. et al.

Fuel injectors for multipoint arrays

Abstract

A combustor dome system includes an annular combustor dome defining a main axis therethrough. The combustor dome includes opposed upstream and downstream faces, wherein the upstream face is configured to face upstream toward a compressor discharge space, wherein the downstream face is configured to face downstream toward a combustor space. The downstream face has a curved cross-sectional profile. A plurality of nozzles extends at least partially through the combustor dome from the upstream face to the downstream face for injection of fuel into the combustor space. A fuel manifold is in fluid communication with the plurality of nozzles.

Inventors: Ryon; Jason A. (Carlisle, IA), Prociw; Lev A. (Johnston, IA), Zink; Gregory A. (Des Moines, IA)

Applicant: Collins Engine Nozzles, Inc. (Des Moines, IA)

Family ID: 1000008820863

Assignee: Collins Engine Nozzles, Inc. (Des Moines, IA)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3018624	12/1961	Bonsall et al.	N/A	N/A
3273343	12/1965	Cretella	N/A	N/A
3453826	12/1968	Hering	N/A	N/A
4374466	12/1982	Sotheran	60/749	F23R 3/16
4763482	12/1987	Wehner	N/A	N/A
4955201	12/1989	Shekleton et al.	N/A	N/A
5195315	12/1992	Holladay	N/A	N/A
5323604	12/1993	Ekstedt et al.	N/A	N/A
5465571	12/1994	Clark	60/800	F23R 3/283
5524430	12/1995	Mazeaud et al.	N/A	N/A
6212870	12/2000	Thompson et al.	N/A	N/A
6250062	12/2000	Lawen, Jr. et al.	N/A	N/A
6360525	12/2001	Senior	60/776	F23R 3/12
6684642	12/2003	Willis et al.	N/A	N/A
7716933	12/2009	Prociw et al.	N/A	N/A
8146837	12/2011	Thomson et al.	N/A	N/A
8313046	12/2011	Short	N/A	N/A
9003802	12/2014	Hernandez et al.	N/A	N/A
9003804	12/2014	Prociw	N/A	N/A
9587833	12/2016	Lee	N/A	N/A
2003/0167771	12/2002	Hayashi et al.	N/A	N/A
2005/0034460	12/2004	McMasters et al.	N/A	N/A
2008/0053102	12/2007	Al-Roub et al.	N/A	N/A
2008/0105237	12/2007	Gandza	N/A	N/A
2012/0198853	12/2011	Hernandez et al.	N/A	N/A
2012/0204575	12/2011	Carlisle	N/A	N/A
2013/0174562	12/2012	Holcomb	60/752	F23R 3/002
2013/0214063	12/2012	Ryon et al.	N/A	N/A
2014/0075949	12/2013	Prociw	N/A	N/A
2014/0318138	12/2013	Bunel et al.	N/A	N/A
2015/0108236	12/2014	Buelow et al.	N/A	N/A
2016/0377293	12/2015	Short et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0495587	12/1991	EP	N/A
2589877	12/2012	EP	N/A
3076082	12/2015	EP	N/A

OTHER PUBLICATIONS

Extended European Search Report dated Aug. 14, 2018, issued during the prosecution of corresponding European Patent Application No. EP 1816329.1. cited by applicant

Primary Examiner: Sung; Gerald L

Attorney, Agent or Firm: Kinney & Lange, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a divisional of U.S. patent application Ser. No. 17/064,766 filed Oct. 7, 2020, which is a continuation of U.S. patent application Ser. No. 15/476,362 filed Mar. 31, 2017, now U.S. Pat. No. 10,859,269 granted Dec. 8, 2020, the contents of which are incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to multipoint injection, and more particularly to multipoint fuel injection such as in gas turbine engines.

2. Description of Related Art

(2) In the early 2000's, the ability of a multipoint lean direct injection (LDI) concept was demonstrated to achieve very low NOx emissions index (EI) levels. The program was successful in demonstrating the ability of the multipoint concept to deal with NOx at high power conditions but was not optimized to perform equally as well at low power conditions such as start, ground idle, and flight idle conditions.

(3) Conventional methods and systems have generally been considered satisfactory for their intended purpose. However, there is still a need in the art for a multipoint fuel injection arrangement that allows for improved performance at high and low power operation while significantly reducing NOx emissions. There also remains a need in the art for such a multipoint fuel injection arrangement that is easy to make and use. This disclosure provides a solution for these needs.

SUMMARY OF THE INVENTION

(4) A combustor dome system includes an annular combustor dome defining a main axis therethrough. The combustor dome includes opposed upstream and downstream faces, wherein the upstream face is configured to face upstream toward a compressor discharge space, wherein the downstream face is configured to face downstream toward a combustor space. The downstream face has a curved cross-sectional profile. A plurality of nozzles extends at least partially through the combustor dome from the upstream face to the downstream face for injection of fuel into the combustor space. A fuel manifold is in fluid communication with the plurality of nozzles.

(5) The annular dome can include an upper cross-section and a lower-cross section, wherein in each of the upper and lower cross-sections, there is symmetry in the curved cross-sectional profile across a cylinder defined about the main axis and extending downstream from downstream face of the

combustor dome. The cylinder can divide the combustor space into an inner space inside the cylinder and an outboard space outside the cylinder, wherein some of the plurality of nozzles are radially inboard relative to the cylinder, and some of nozzles in the plurality of nozzles are radially outboard relative to the cylinder. The nozzles of the plurality of nozzles that are radially inboard of the cylinder can be angled outward toward the cylinder. The nozzles of the plurality of nozzles that are radially outboard of the cylinder can be angled inward toward the cylinder.

(6) Each nozzle in the plurality of nozzles can include a nozzle assembly defining at least one of a fuel circuit and an air circuit therethrough, wherein the nozzle assembly extends along an injection axis. A plurality of supports can extend from the nozzle assembly to engage the combustor dome, the supports being circumferentially spaced apart from one another about the injection axis, wherein each support in the plurality of supports extends along a respective support axis. The support axes are circumferentially spaced apart around an insertion axis along which the nozzle assembly can be inserted during assembly into a combustor dome that is oblique relative to the injection axis

(7) Each support includes opposing side faces that are aligned with the insertion axis. Each support in the plurality of supports can be aerodynamically aligned with the injection axis. At least one of the supports can be swept on a non-perpendicular angle relative to the injection axis. At least one of the supports can include a fuel feed passage fluidly connected with a fuel circuit of the nozzle assembly for fluid communication of fuel from a manifold into the fuel circuit. The combustor dome can define an outer air flow passage therethrough for each nozzle aligned with the injection axis thereof, wherein the nozzle assembly is suspended in the outer air flow passage of the combustor dome with an outer air circuit defined through the outer air flow passage outboard of the nozzle assembly. The fuel manifold can be supported by fuel conduits that provide fluid communication from the fuel manifold to the plurality of nozzles. The manifold can hold the plurality of nozzles in place axially and forces them up against the dome to maintain a constant gap.

(8) The dome can be configured to provide cooling due to air flowing therealong. The dome can be configured to provide partial channel heat transfer for cooling the dome and maintaining metering of air flow through an outer air circuit defined between the dome and each of the plurality of nozzles. The supports can provide alignment and concentricity for the nozzles.

(9) A method of assembling a combustor dome system includes assembling a nozzle into a combustor dome that defines a main axis, wherein assembling includes inserting the nozzle into an outer air flow passage defined through the combustor dome in a direction aligned parallel with the main axis, wherein the nozzle itself is aligned along an injection axis that is oblique to the main axis during insertion.

(10) These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

(2) FIG. 1 is a schematic partially cross-sectional side elevation view of an exemplary embodiment of a gas turbine engine constructed in accordance with the present disclosure, showing upper and lower cross-sections of an annular combustor dome;

- (3) FIG. 2 is a cross-sectional perspective view of the annular combustor dome of FIG. 1, showing the curvature of the downstream surface of the combustor dome and the nozzle injection angles;
- (4) FIG. 3 is a cross-sectional side elevation view of the upper cross-section of the annular combustor dome of FIG. 1, showing inner and outer nozzles and the fuel manifold;
- (5) FIG. 4 is an upstream end elevation view of the combustor dome of FIG. 1, showing the nozzles and upstream surface of the combustor dome viewed from the compressor side;
- (6) FIG. 5 is an upstream end elevation view of the combustor dome of FIG. 4, showing the manifold; and
- (7) FIGS. 6 and 7 are perspective and cross-sectional perspective views of one of the nozzles of the system of FIG. 5, showing the supports and fuel and air circuits.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an exemplary embodiment of a system in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. Other embodiments of systems in accordance with the disclosure, or aspects thereof, are provided in FIGS. 2-7, as will be described. The systems and methods described herein can be used for multipoint injection, such as in gas turbine engines.

(9) Gas turbine engine **10** includes a compressor section **12** for compressing air to an elevated pressure, a combustor section **14** for burning fuel in the compressed air, and a turbine section **16** for extracting energy from the heated, compressed air to generate shaft work and or exhaust thrust. Combustor dome system **100** is the portion of the combustor section **14** that separates the compressor space **18** of compressor **12** from the combustor space **20** of combustor section **14**, and introduces fuel and air into the combustor space **20** for combustion.

(10) With reference now to FIG. 2, combustor dome system **100** includes an annular combustor dome **102** defining a main axis A therethrough. The combustor dome **102** includes opposed upstream and downstream faces **104** and **106**, wherein the upstream face **104** is configured to face upstream toward a compressor discharge space **18**, wherein the downstream face is configured to face downstream into combustor space **20**. The downstream face **106** has a curved cross-sectional profile **110**, which can be seen in both the upper and lower cross-sections **112** and **114** showing in FIG. 2. A plurality of nozzles **116** extends through the combustor dome **102** from the upstream face **104** to the downstream face **106** for injection of fuel into the combustor space **20**. A fuel manifold **118** is in fluid communication with the plurality of nozzles **116** for delivery of the fuel from a fuel source to the nozzles **116**.

(11) There can be symmetry in the curved cross-sectional profile (shown in both the upper and lower cross-sections **112** and **114** in FIG. 2, but present in all cross-sections) across a cylinder **120** defined about the main axis A and extending downstream from downstream face **106** of the combustor dome **102** (note that in FIG. 3, cylinder **120** appears as a line due to the cross-section and is not to be mistaken with axis A shown in FIG. 2). Those skilled in the art will appreciate that cylinder **120** is not an actual component, but is a geometric reference. The cylinder **120** can divide the combustor space into an inner space **122** inside the cylinder **120** and an outboard space **124** outside the cylinder **120**. Some of the plurality of nozzles **116** are radially inboard relative to the cylinder **120** (an inner set of nozzles **116**), and some of nozzles **116** are radially outboard relative to the cylinder **120** (an outer set of nozzles **116**). The nozzles **116** that are radially outboard of the cylinder **120** are angled inward toward the cylinder on angle $\theta 1$. The nozzles **116** that are radially inboard of the cylinder **120** are angled outward toward the cylinder **120** on angle $\theta 2$. There is also a central set of nozzles **116** radially between the inner and outer sets that are aligned with cylinder **120** and parallel to main axis A. Thus nozzles **116** conform in their injection angle to the curved cross-sectional profile **110** of the downstream surface **106** of combustor dome **102**.

(12) Referring now to FIG. 3, each nozzle **116** includes a nozzle assembly **126** defining a fuel

circuit **128** and an air circuit **130** therethrough, wherein the nozzle assembly **126** extends along an injection axis I, angled on converging/diverging angles θ_1 , θ_2 , or aligned with cylinder **120** as described above. The nozzle assembly **126** includes inner and outer fuel circuit components **154** and **156**, a receptacle **158**, and an air swirler **160** described below. Two or more supports **132** extend from each nozzle assembly **126** to engage the combustor dome **102** to support each nozzle **116** in the combustor dome **102**. The three supports **132** are circumferentially spaced apart 120° from one another about the injection axis I, as shown in FIG. 4. Each support **132** extends along a respective support axis S, and it is these support axes S that are spaced apart circumferentially, 120° in this example. Those skilled in the art will readily appreciate that any other suitable angular spacing can be used between the supports **132**, and it is advantageous structurally if the supports **132** are not collinear. Each support **132** in the plurality of supports includes side faces **134** shown in FIGS. 6 and 7. The side faces **134** of each support **132** are circumferentially spaced apart from the side faces of other supports **132** around an insertion axis i that is oblique relative to the injection axis I and parallel to the main axis A and cylinder **120**. Each side face **134** is aligned with the respective insertion axis i so that nozzles **116** can be inserted into combustor dome **102** along insertion axes i that are all aligned parallel with main axis A. As shown in FIG. 4 each support **132** in the plurality of supports is aerodynamically aligned with the injection axis I, i.e. as vanes with no turning or swirl angle to reduce or minimize the wake caused in air flow past supports **132**. Supports **132** of each nozzle **116** can be swept forward or backwards as needed on a non-perpendicular angle α relative to the injection axis I to provide the insertion axis i parallel to main axis A and cylinder **110**. At least one of the supports **132** in each injector **116** can include a fuel feed passage **136** fluidly connected with the fuel circuit **128** of the nozzle assembly **126** for fluid communication of fuel from the manifold **118** into the fuel circuit **128**.

(13) The groove side surfaces **135** shown in FIG. 4 are also aligned parallel to the support axis S, shown in FIG. 6 such that the groove is co-linear with the support axis S. The groove side surfaces **135** and corresponding side faces **134** of supports **132** are also aligned parallel with the insertion axis i. The tighter the gap between the support side wall **134** (shown in FIG. 6) and groove side wall **135** (shown in FIG. 4) the tighter constrained the component supports **132** will be and this gap can be minimized as to the point that assembly is still feasible. The groove end walls **135** extend beyond the end of each support **132** in the direction of support axis S. This still allows the grooves and the supports **132** to grow/shrink radially inwards/outwards from the insertion axis i.

(14) With reference again to FIG. 3, the combustor dome **102** defines an outer air flow passage **138** therethrough for each nozzle **116** that is aligned with the injection axis I thereof. The nozzle assembly **126** is suspended in the outer air flow passage **138** of the combustor dome **102** with an outer air circuit defined through the outer air flow passage **138** outboard of the nozzle assembly **102**. The fuel manifold **118** supports by the fuel conduits of feed passages **136** the nozzles **116** and holds the nozzles **116** in place axially, forcing them against the combustor dome **102** to maintain a consistent gap in each outer air flow passage **138** between each nozzle **116** and the combustor plate **102**. The outer air flow passages **138** can provide improved cooling of the combustor dome **102** compared to traditional arrangements. The entire assembly of nozzles **116** and manifold **118** can thus be inserted into combustor dome **102** along the main axis A despite the various injection angles I of the different nozzles **116** during insertion.

(15) Referring to FIGS. 2 and 3, manifold **118** includes three fuel circuits **140**, **142**, and **144**, connecting from main feed arms **150**, through outer manifold ring **146** and into arms **148** for delivery of separately manifolded fuel to each of the inner, outer, and central sets of nozzles **116**. Thus all three sets of nozzles **116** can be staged for separate operation. FIG. 5 shows the complete outer and inner rings **146** and **152**, as well as all of the arms **148**, each of which delivers fuel to one each of an inner nozzle **116**, central nozzle **116**, and outer nozzle **116**. Feed arms **150** can provide support for manifold **118** in engine **10** of FIG. 1. Inner ring **152** of manifold **118** is devoid of fuel circuits and is circumferentially segmented to accommodate thermal expansion and contraction.

(16) With reference now to FIG. 7, assembly of each nozzle **116** can include brazing inner and outer fuel circuit components **154** and **156** together, and the receptacle **158** for conduit **136** can be drilled. The assembled inner and outer fuel circuits **154** and **156** can be loaded from the downstream side into nozzle assembly **126**, which can form a heat shield and can be made of a ceramic material, metallic material, or any other suitable material, and can be unitary with supports **132**. The fuel tube of conduit **136** can be assembled into fluid communication with fuel circuit **128**, and an inner air swirler **160** can be assembled into air circuit **130**. There are thus two air circuits, an inner (**130**) and outer (**138** shown in FIGS. 3 and 4) air circuit sandwiching the fuel circuit **128** for each nozzle **116**.

(17) The dome **102** is configured to provide cooling due to air flowing therealong. The dome **102** is configured to provide partial channel heat transfer for cooling the dome **102** and maintaining metering of air flow through the outer air circuit defined between the dome **102** and each of the plurality of nozzles **116**. The supports **132** can provide alignment and concentricity for the nozzles **116**.

(18) Those skilled in the art will readily appreciate that while shown and described with one fuel circuit and one air circuit in each nozzle, any suitable number of fuel and air circuits can be used without departing from the scope of this disclosure. Those skilled in the art will also readily appreciate that while three fuel circuits corresponding to inner, outer, and central nozzles, any suitable number of circuits or sets of nozzles can be used without departing from the scope of this disclosure. For example, five rings of nozzles can be included, wherein an intermediate set of nozzles is provided with injection angles conforming to the curvature of the combustor dome between the central set of nozzles and each of the inner and outer sets of nozzles.

(19) The arrangements disclosed herein provide controlled spacing between nozzles and combustor domes. These disclosed arrangements increase fuel/air mixing efficiency compared to conventional arrangements by using air which would otherwise bypass the nozzle directly into the mixing path of the nozzle. There is also a reduced part count for reduced cost and weight compared to conventional arrangements. The arrangements disclosed herein can be suitable for ceramic and ceramic matrix material constructions, and can improve heat transfer to the dome plate relative to conventional arrangements.

(20) The methods and systems of the present disclosure, as described above and shown in the drawings, provide for combustor dome systems with superior properties including improved multipoint injection and ease of assembly relative to conventional systems. While the apparatus and methods of the subject disclosure have been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A combustor dome system comprising: an annular combustor dome defining a main axis therethrough, the combustor dome including opposed upstream and downstream faces, wherein the upstream face is configured to face upstream toward a compressor discharge space, wherein the downstream face is configured to face downstream toward a combustor space, and wherein the downstream face has a curved cross-sectional profile; a plurality of nozzles extending at least partially through the combustor dome from the upstream face to the downstream face for injection of fuel into the combustor space; and a nozzle assembly defining a fuel circuit therethrough, wherein the nozzle assembly extends along an injection axis; a plurality of supports extending from the nozzle assembly to engage the combustor dome wherein at least one of the supports includes a fuel feed passage fluidly connected with the fuel circuit of the nozzle assembly for fluid communication of fuel from a fuel manifold into the fuel circuit.

2. The system as recited in claim 1, wherein the annular dome includes an upper cross-section and

a lower-cross section, wherein in each of the upper and lower cross-sections, there is symmetry in the curved cross-sectional profile across a cylinder defined about the main axis and extending downstream from the downstream face of the combustor dome.

3. The system as recited in claim 1, wherein a cylinder defined about the main axis and extending downstream from downstream face of the combustor dome divides the combustor space into an inner space inside the cylinder and an outboard space outside the cylinder, wherein some of the plurality of nozzles are radially inboard relative to the cylinder, and some of the plurality of nozzles are radially outboard relative to the cylinder.

4. The system as recited in claim 3, wherein the nozzles of the plurality of nozzles that are radially inboard of the cylinder are angled outward toward the cylinder.

5. The system as recited in claim 3, wherein the nozzles of the plurality of nozzles that are radially outboard of the cylinder are angled inward toward the cylinder.

6. The system as recited in claim 1, wherein each support in the plurality of supports includes a side face, wherein the side faces are circumferentially spaced apart around an insertion axis that is oblique relative to the injection axis and parallel to the main axis.

7. The system as recited in claim 6, wherein each side face is aligned with the insertion axis.

8. The system as recited in claim 6, wherein each support in the plurality of supports is aerodynamically aligned with the injection axis.

9. The system as recited in claim 6, wherein at least one of the supports is swept on a non-perpendicular angle relative to the injection axis.

10. The system as recited in claim 6, wherein the combustor dome defines an outer air flow passage therethrough for each nozzle aligned with the injection axis thereof, wherein the nozzle assembly is suspended in an outer air flow passage of the combustor dome with an outer air circuit defined through the outer air flow passage outboard of the nozzle assembly.

11. The system of claim 6, wherein the supports provide alignment and concentricity for the nozzles.

12. The system of claim 6, wherein each support includes an end face that faces radially outward relative to the insertion axis, wherein each end face is contained in a respective plane that is parallel to the insertion axis and oblique to the injection axis.

13. The system as recited in claim 1, wherein the fuel manifold is supported by fuel conduits that provide fluid communication from the fuel manifold to the plurality of nozzles, wherein the manifold holds the plurality of nozzles in place axially and forces the plurality of nozzles up against the dome to maintain a constant gap.

14. A combustor dome system comprising: an annular combustor dome defining a main axis therethrough, the combustor dome including opposed upstream and downstream faces, wherein the upstream face is configured to face upstream toward a compressor discharge space, wherein the downstream face is configured to face downstream toward a combustor space, and wherein the downstream face has a curved cross-sectional profile; a plurality of nozzles extending at least partially through the combustor dome from the upstream face to the downstream face for injection of fuel into the combustor space, each nozzle in the plurality of nozzles further comprising: a nozzle assembly defining at least one of a fuel circuit and an air circuit therethrough, wherein the nozzle assembly extends along an injection axis; and a plurality of supports extending from the nozzle assembly to engage the combustor dome, the supports being circumferentially spaced apart from one another about the injection axis, wherein each support in the plurality of supports includes a side face, wherein the side faces are circumferentially spaced apart around an insertion axis that is oblique relative to the injection axis and parallel to the main axis wherein at least one of the supports includes a fuel feed passage fluidly connected with a fuel circuit of the nozzle assembly for fluid communication of fuel from a manifold into the fuel circuit; and a fuel manifold in fluid communication with the plurality of nozzles.

15. The system as recited in claim 14, wherein the combustor dome defines an outer air flow

passage therethrough for each nozzle aligned with the injection axis thereof, wherein the nozzle assembly is suspended in the outer air flow passage of the combustor dome with an outer second air circuit defined through the outer air flow passage outboard of the nozzle assembly.

16. The system of claim 14, wherein the supports provide alignment and concentricity for the nozzles.

17. The system of claim 14, wherein each support includes an end face that faces radially outward relative to the insertion axis, wherein each end face is contained in a respective plane that is parallel to the insertion axis and oblique to the injection axis.

18. The system as recited in claim 14, wherein each side face is aligned with the insertion axis.

19. The system as recited in claim 14, wherein each support in the plurality of supports is aerodynamically aligned with the injection axis.

20. The system as recited in claim 1, wherein the fuel manifold is supported by fuel conduits that provide fluid communication from the fuel manifold to the plurality of nozzles, wherein the manifold holds the plurality of nozzles in place axially and forces the plurality of nozzles up against the dome to maintain a constant gap.
